

ACR15030

## 1,1,1-Trichloro-2-methyl-2-propanol hemihydrate

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** 1,1,1-三氯-2-甲基-2-丙醇半水合物  
1,1,1-Trichloro-2-methyl-2-propanol hemihydrate

**Cat No. :** 150300000; 150301000; 150305000; 150300025  
**Synonyms** beta,beta,beta-Trichloro-tert-butyl alcohol  
**CAS No** 6001-64-5  
**Molecular Formula** C4 H7 Cl3 O . 0.5 H2 O

**Supplier**

**UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road,  
Loughborough, Leicestershire LE11 5RG, United Kingdom

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium

**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Solid

**Appearance**  
White

**Odor**  
Strong

#### Emergency Overview

Causes skin irritation. Causes serious eye irritation. Hygroscopic. Harmful if swallowed.

#### Classification of the substance or mixture

|                                                    |            |
|----------------------------------------------------|------------|
| Acute Oral Toxicity                                | Category 4 |
| Skin Corrosion/Irritation                          | Category 2 |
| Serious Eye Damage/Eye Irritation                  | Category 2 |
| Specific target organ toxicity - (single exposure) | Category 3 |

#### Label Elements



# SAFETY DATA SHEET

## 1,1,1-Trichloro-2-methyl-2-propanol hemihydrate

**Signal Word****Warning****Hazard Statements**

H315 - Causes skin irritation

H319 - Causes serious eye irritation

H302 - Harmful if swallowed

**Precautionary Statements****Prevention**

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P312 - Call a POISON CENTER or doctor if you feel unwell

P330 - Rinse mouth

P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

Hygroscopic.

**Health Hazards**

Causes skin irritation. Causes serious eye irritation. Harmful if swallowed.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants.

This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component                                          | CAS No    | Weight % |
|----------------------------------------------------|-----------|----------|
| 2-Propanol, 1,1,1-trichloro-2-methyl-, hemihydrate | 6001-64-5 | >95      |

**SECTION 4. FIRST AID MEASURES****Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Get medical attention. Wash off immediately with plenty of water for at least 15 minutes.

**Inhalation**

Remove to fresh air. Get medical attention. If not breathing, give artificial respiration.

**Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

**Most important symptoms and effects**

No information available.

**Self-Protection of the First Aider**

# SAFETY DATA SHEET

## 1,1,1-Trichloro-2-methyl-2-propanol hemihydrate

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Keep product and empty container away from heat and sources of ignition. Thermal decomposition can lead to release of irritating gases and vapors.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation. Avoid contact with skin, eyes or clothing. Keep people away from and upwind of spill/leak.

**Environmental Precautions**

See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Avoid contact with skin and eyes. Do not breathe dust. Do not ingest. If swallowed then seek immediate medical assistance. Ensure adequate ventilation. Wear personal protective equipment/face protection. Avoid dust formation.

**Storage**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and

# SAFETY DATA SHEET

## 1,1,1-Trichloro-2-methyl-2-propanol hemihydrate

inhalable dust

### Exposure Controls

#### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

#### Personal protective equipment

**Eye Protection** Wear safety glasses with side shields (or goggles) (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection** No protective equipment is needed under normal use conditions.

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

**Small scale/Laboratory use** Maintain adequate ventilation

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                 |                               |                                          |
|---------------------------------|-------------------------------|------------------------------------------|
| <b>Appearance</b>               | White                         |                                          |
| <b>Physical State</b>           | Solid                         |                                          |
| <b>Odor</b>                     | Strong                        |                                          |
| <b>Odor Threshold</b>           | No data available             |                                          |
| <b>pH</b>                       | 4.5-6 (@ 20 )                 | 7.7 g/L (20°C)                           |
| <b>Melting Point/Range</b>      | 75 - 79 °C / 167 - 174.2 °F   |                                          |
| <b>Softening Point</b>          | No data available             |                                          |
| <b>Boiling Point/Range</b>      | 173 - 175 °C / 343.4 - 347 °F | @ 760 mmHg                               |
| <b>Flash Point</b>              | 100 °C / 212 °F               | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>         | Not applicable                | Solid                                    |
| <b>Flammability (solid,gas)</b> | No information available      |                                          |
| <b>Explosion Limits</b>         | No data available             |                                          |
| <b>Vapor Pressure</b>           | No data available             |                                          |
| <b>Vapor Density</b>            | Not applicable                | Solid                                    |

## 1,1,1-Trichloro-2-methyl-2-propanol hemihydrate

|                                                    |                          |       |
|----------------------------------------------------|--------------------------|-------|
| Specific Gravity / Density                         | No data available        |       |
| Bulk Density                                       | No data available        |       |
| Water Solubility                                   | No information available |       |
| Solubility in other solvents                       | No information available |       |
| Partition Coefficient (n-octanol/water)            |                          |       |
| Component                                          | log Pow                  |       |
| 2-Propanol, 1,1,1-trichloro-2-methyl-, hemihydrate | 2.03                     |       |
| Autoignition Temperature                           | No data available        |       |
| Decomposition Temperature                          | > 167°C                  |       |
| Viscosity                                          | Not applicable           | Solid |
| Explosive Properties                               | No information available |       |
| Oxidizing Properties                               | No information available |       |
| Molecular Formula                                  | C4 H7 Cl3 O . 0.5 H2 O   |       |
| Molecular Weight                                   | 186.47                   |       |

## SECTION 10. STABILITY AND REACTIVITY

|                          |                                                                          |
|--------------------------|--------------------------------------------------------------------------|
| Stability                | Hygroscopic.                                                             |
| Hazardous Reactions      | No information available.                                                |
| Hazardous Polymerization | No information available.                                                |
| Conditions to Avoid      | Excess heat. Incompatible products. Exposure to moist air or water.      |
| Materials to avoid       | Acids. Strong oxidizing agents. Acid anhydrides. Acid chlorides. Metals. |

**Hazardous Decomposition Products** Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Chlorine. Hydrogen chloride gas.

## SECTION 11. TOXICOLOGICAL INFORMATION

## Product Information

## (a) acute toxicity;

| Component                                          | LD50 Oral       | LD50 Dermal          | LC50 Inhalation |
|----------------------------------------------------|-----------------|----------------------|-----------------|
| 2-Propanol, 1,1,1-trichloro-2-methyl-, hemihydrate | 510 mg/kg (Rat) | >2000 mg/kg (Rabbit) |                 |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

## (d) respiratory or skin sensitization;

|             |                   |
|-------------|-------------------|
| Respiratory | No data available |
| Skin        | No data available |

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

# SAFETY DATA SHEET

## 1,1,1-Trichloro-2-methyl-2-propanol hemihydrate

|                                            |                                                                |
|--------------------------------------------|----------------------------------------------------------------|
| (h) STOT-single exposure;                  | Category 3                                                     |
| (i) STOT-repeated exposure;                | No data available                                              |
| Target Organs                              | No information available.                                      |
| (j) aspiration hazard;                     | Not applicable<br>Solid                                        |
| Other Adverse Effects                      | The toxicological properties have not been fully investigated. |
| Symptoms / effects, both acute and delayed | No information available                                       |

### SECTION 12. ECOLOGICAL INFORMATION

|                     |                                                                                                                             |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Ecotoxicity effects | Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants. |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------|

| Component                                          | Freshwater Fish                                          | Water Flea | Freshwater Algae | Microtox |
|----------------------------------------------------|----------------------------------------------------------|------------|------------------|----------|
| 2-Propanol, 1,1,1-trichloro-2-methyl-, hemihydrate | LC50: = 135 mg/L, 96h flow-through (Pimephales promelas) |            |                  |          |

|                                              |                                                      |
|----------------------------------------------|------------------------------------------------------|
| Persistence and Degradability<br>Persistence | No information available<br>Persistence is unlikely. |
|----------------------------------------------|------------------------------------------------------|

|                           |                             |
|---------------------------|-----------------------------|
| Bioaccumulative Potential | Bioaccumulation is unlikely |
|---------------------------|-----------------------------|

| Component                                          | log Pow | Bioconcentration factor (BCF) |
|----------------------------------------------------|---------|-------------------------------|
| 2-Propanol, 1,1,1-trichloro-2-methyl-, hemihydrate | 2.03    | No data available             |

|                  |                          |
|------------------|--------------------------|
| Mobility in soil | No information available |
|------------------|--------------------------|

|                                 |                                                                           |
|---------------------------------|---------------------------------------------------------------------------|
| Endocrine Disruptor Information | This product does not contain any known or suspected endocrine disruptors |
| Persistent Organic Pollutant    | This product does not contain any known or suspected substance            |
| Ozone Depletion Potential       | This product does not contain any known or suspected substance            |

### SECTION 13. DISPOSAL CONSIDERATIONS

|                                     |                                                                                                                                                                        |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Waste from Residues/Unused Products | Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations. |
| Contaminated Packaging              | Dispose of this container to hazardous or special waste collection point.                                                                                              |
| Other Information                   | Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.                                          |

### SECTION 14. TRANSPORT INFORMATION

|                         |               |
|-------------------------|---------------|
| Road and Rail Transport | Not Regulated |
|-------------------------|---------------|

# SAFETY DATA SHEET

## 1,1,1-Trichloro-2-methyl-2-propanol hemihydrate

**IMDG/IMO** Not regulated**IATA** Not regulated**Special Precautions for User** No special precautions required

### SECTION 15. REGULATORY INFORMATION

#### International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component                                          | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL |
|----------------------------------------------------|-----------------------------------------------------|-----------------------------------------|------|-------|--------|------|-----|-------|------|------|------|------|
| 2-Propanol, 1,1,1-trichloro-2-methyl-, hemihydrate | -                                                   | -                                       | X    | X     | -      | -    | -   | -     | -    |      | X    | -    |

#### National Regulations

### SECTION 16. OTHER INFORMATION

**Revision Date** 06-Apr-2024  
**Revision Summary** Not applicable.

#### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

#### Legend

**CAS** - Chemical Abstracts Service**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances**PICCS** - Philippines Inventory of Chemicals and Chemical Substances**IECSC** - Chinese Inventory of Existing Chemical Substances**KECL** - Korean Existing and Evaluated Chemical Substances**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List**ENCS** - Japanese Existing and New Chemical Substances**AICS** - Australian Inventory of Chemical Substances**NZIoC** - New Zealand Inventory of Chemicals**WEL** - Workplace Exposure Limit**ACGIH** - American Conference of Governmental Industrial Hygienists**DNEL** - Derived No Effect Level**RPE** - Respiratory Protective Equipment**LC50** - Lethal Concentration 50%**NOEC** - No Observed Effect Concentration**PBT** - Persistent, Bioaccumulative, Toxic**TWA** - Time Weighted Average**IARC** - International Agency for Research on Cancer**PNEC** - Predicted No Effect Concentration**LD50** - Lethal Dose 50%**EC50** - Effective Concentration 50%**POW** - Partition coefficient Octanol:Water**vPvB** - very Persistent, very Bioaccumulative

# SAFETY DATA SHEET

## 1,1,1-Trichloro-2-methyl-2-propanol hemihydrate

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**